DSTRA-GNN Results Summary

Short report for supervisor

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1 Purpose and Evaluation Protocol

This report summarizes the final experimental results for DSTRA-GNN, a dynamic spatio-temporal route-aware graph neural network for per-vehicle ETA prediction. The model represents traffic as a dynamic graph with static road topology, vehicle nodes, interaction edges, explicit remaining-route features, temporal context, and a Mixture-of-Experts ETA head.

The evaluation uses two graph datasets and one trajectory-native dataset for external baselines. SUMO is the controlled simulation domain. After detecting a rare extreme-duration simulator tail, all reported SUMO statistics and broad-population results use a p95-filtered corpus that removes trips longer than 2,587 s, retaining 798,919 trips. Porto-G is the graph-converted Porto taxi dataset. Porto-T is the trajectory-native filtered Porto dataset used for comparison with DeepTTE, MetaTTE, and other trajectory baselines.

Protocol. All datasets use chronological train/validation/test splits with no leakage across splits. SUMO uses two weeks for training, one week for validation, and one week for testing. Porto uses the shared calendar split: train 2013-07-01–2014-02-28, validation 2014-03-01–2014-04-01, and test 2014-05-01–2014-07-01. Graph models use 30-s snapshots, 30-snapshot windows, and one ETA prediction per trip start. Porto-T and Porto-G are evaluated in the 315–945 s MetaTTE-style protocol range; SUMO is reported both in that shared range and on the broader p95-filtered population. Model selection is by validation MAE. A3–A6 are reported as mean ± standard deviation over seeds 42/43/44; A1–A2 use seed 42. Headline metrics are test-set MAE, RMSE, and MAPE after inverting the normalized log target to seconds.

2 Ablation Variants and Results

The six DSTRA-GNN variants isolate the effect of dynamic edges, route awareness, and temporal context. A6 is the full model with all three components enabled.

Table 1: Ablation variants and enabled components.

Variant	Dyn. edges	Route feat.	Temporal GRU
A1 base_graph	–	–	–
A2 dynamic_graph	✓	–	–
A3 route_aware_graph	✓	✓	–
A4 temporal_base	–	–	✓
A5 temporal_dynamic	✓	–	✓
A6 temporal_route_aware	✓	✓	✓

Table 2: Full DSTRA-GNN ablation results. A3–A6 report mean ± std across seeds 42/43/44; A1–A2 use seed 42 only.

	Porto-G protocol			SUMO protocol			SUMO p95		
Variant	**MAE**	**RMSE**	**MAPE**	**MAE**	**RMSE**	**MAPE**	**MAE**	**RMSE**	**MAPE**
A1 static only	93.29	122.04	17.28	80.10	211.86	12.03	78.33	235.56	25.13
A2 +dyn. edges	91.76	120.63	16.88	74.96	135.63	11.59	77.56	244.34	23.54
A3 +route+dyn.	74.81±0.52	100.30±0.88	13.96	66.81±0.64	110.23±0.93	10.06	76.22±1.42	218.98±2.01	22.34
A4 +temp.	90.00±0.66	118.67±1.01	17.05	76.29±0.73	121.26±1.08	12.05	80.38±1.23	232.16±2.12	24.35
A5 +temp.+dyn.	87.99±0.61	107.83±0.94	16.76	64.08±0.66	97.38±1.01	9.76	75.81±1.82	219.22±2.19	22.22
A6 full model	**68.05±0.48**	**90.01±0.86**	**13.20**	**54.75±0.71**	**82.72±0.96**	**8.33**	**72.44±1.31**	**214.50±2.92**	**19.71**

The protocol-range results give the cleanest cross-domain comparison because Porto-G and Porto-T are duration-matched to the 315–945 s MetaTTE-style range, while SUMO can be evaluated on the same range. In this shared range, A6 gives 27% MAE reduction on Porto-G and 32% on SUMO relative to A1. The broader SUMO p95 view includes trips up to 2,587 s; even there, A6 remains the best variant with 72.44 s MAE. This supports the main architectural claim: route information and temporal context are complementary rather than interchangeable.

3 Porto Baseline Positioning

Porto is the main setting where DSTRA-GNN can be compared with external trajectory-native baselines under a shared chronological split. The comparison should be read as a modality comparison: DSTRA-GNN uses graph snapshots and explicit remaining-route state on Porto-G, while DeepTTE, MetaTTE, WDR, and STNN use GPS trajectory sequences on Porto-T.

Table 3: Porto test-set comparison with strongest external baselines.

Model	**Input**	**MAE**	**RMSE**	**MAPE**
GBM (LightGBM)	Porto-T trajectory/tabular	86.41 s	113.21 s	15.00%
DeepTTE	Porto-T trajectory	68.59 s	90.68 s	**12.13%**
MetaTTE	Porto-T trajectory	69.46 s	91.74 s	12.34%
WDR	Porto-T trajectory	72.11 s	94.60 s	12.81%
STNN	Porto-T trajectory	74.15 s	97.27 s	13.05%
DCRNN	Porto-G graph	131.32 s	172.82 s	28.95%
DSTRA-GNN A6	**Porto-G graph**	**68.05 s**	**90.01 s**	13.20%

DSTRA-GNN is in the same accuracy band as the strongest trajectory-native deep baselines. It has slightly better MAE and RMSE than DeepTTE and MetaTTE, while DeepTTE retains the best MAPE. The important result is therefore not a large separation from trajectory models, but that a graph-native, route-aware formulation reaches comparable performance while operating on dynamic graph snapshots.

4 Interpretation

- **Route intent is the strongest non-temporal signal.** Adding route features on top of dynamic edges reduces MAE substantially in both domains, especially on Porto-G where sparse taxi observations make local traffic interactions less informative by themselves.
- **Temporal context provides additional gains.** The full A6 model improves over both the route-aware non-temporal model and the temporal dynamic model, showing that route state and temporal aggregation capture different information.
- **Dynamic edges help more in dense traffic.** SUMO has roughly 288 active vehicles per snapshot after p95 filtering, while Porto-G has about 8. Dynamic vehicle–vehicle and vehicle–junction edges are therefore more useful in SUMO than in the sparse Porto taxi graph.
- **The p95 SUMO correction improves scientific validity.** Removing trips above 2,587 s avoids letting a rare simulator tail dominate aggregate metrics, while keeping nearly 95% of the generated trips. The retained SUMO population is still broader than the Porto protocol range and is used consistently throughout the final paper.

5 Bottom Line

The final results support the paper's central claim: DSTRA-GNN provides an open, route-aware dynamic graph formulation for ETA prediction that combines static road structure, vehicle dynamics, temporal context, and remaining-route intent. On both SUMO and Porto-G, the full route-aware temporal model is the best graph variant. On Porto, it reaches the accuracy range of strong trajectory-native deep baselines, demonstrating that graph-native ETA prediction can be competitive when explicit route state is available.